

Elastic-Link for Binarized Neural Network

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Abstract

Recent work has shown that Binarized Neural Networks (BNNs) are able to greatly reduce computational costs and memory footprints, facilitating model deployment on resource-constrained devices. However, in comparison to their full-precision counterparts, BNNs suffer from severe accuracy degradation. Research aiming to reduce this accuracy gap has thus far largely focused on specific network architectures with few or no 1×1 convolutional layers, for which standard binarization methods do not work well. Because 1×1 convolutions are common in the design of modern architectures (e.g. GoogleNet, ResNet, DenseNet), it is crucial to develop a method to binarize them effectively for BNNs to be more widely adopted. In this work, we propose an “Elastic-Link” (EL) module to enrich information flow within a BNN by adaptively adding real-valued input features to the subsequent convolutional output features. The proposed EL module is easily implemented and can be used in conjunction with other methods for BNNs. We demonstrate that adding EL to BNNs produces a significant improvement on the challenging large-scale ImageNet dataset. For example, we raise the top-1 accuracy of binarized ResNet26 from 57.9% to 64.0%. EL also aids convergence in the training of binarized MobileNet, for which a top-1 accuracy of 56.4% is achieved. Finally, with the integration of ReActNet, it yields a new state-of-the-art result of 71.9% top-1 accuracy.

Introduction

Convolutional Neural Networks (CNNs) have led to a series of breakthroughs for a variety of visual tasks (Krizhevsky, Sutskever, and Hinton 2012; Long, Shelhamer, and Darrell 2014; Ren et al. 2015; Toshev and Szegedy 2014; Zhu et al. 2016b). However, the challenge of resource constraints in terms of latency and memory storage is often faced when deploying CNNs on mobile or embedded devices. Previous work (Cai et al. 2017; Jacob et al. 2018; McKinstry et al. 2019; Jain et al. 2020; Shuang Wu and Shi 2016; Sung, Shin, and Hwang 2015; Yang et al. 2019; Zhou et al. 2016) has demonstrated that quantizing the real-valued weights and activations of CNNs into low-precision representations can reduce memory footprint while still achieving good performances. This class of methods allows fixed-point arithmetic

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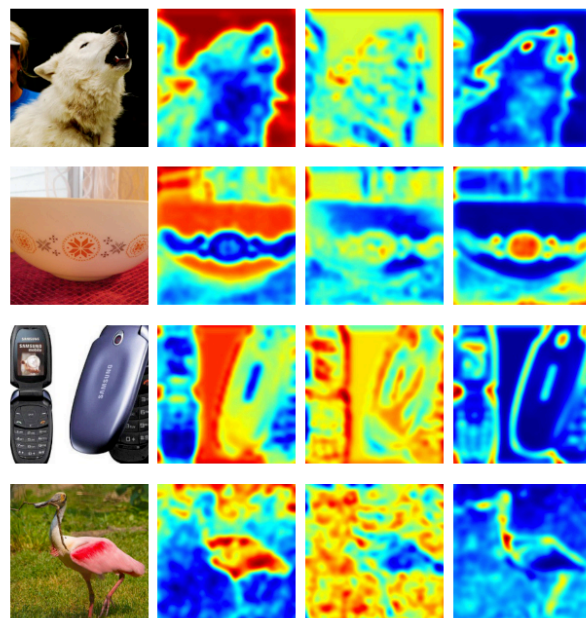


Figure 1: Example images illustrating the same features on full-precision ResNet26 (2nd column), Bi-Real-ResNet26 (3rd column) and our proposed EL-ResNet26 (4th column).

to be applied, which substantially accelerates inference and reduces energy costs. Taken to an extreme, both the weights and the activations can be represented with binary tensors $\{-1, +1\}$. Such networks are termed Binarized Neural Networks (BNNs) (Courbariaux et al. 2016). In BNNs, arithmetic operations for convolutions can be replaced by the more efficient *xnor* and *bitcount* operations.

However, BNNs suffer from significant accuracy degradation as a result of information loss at each binarized layer. To conduct binarization, a real-valued signal is passed through the *Sign* activation, which eliminates the signal’s amplitude and retains only its sign information. Because this process is irreversible, information loss increases with each layer of the BNN. Therefore, a central challenge to improving BNN accuracy is the reduction of this information loss.